

FitBuddy Project Development Phase

Technical Execution, Frontend Stack, Feature Integration, and Code Governance

01. Project Development Overview

The project development phase converts the finalized design specifications and architectural blueprints into a fully functional **FitBuddy** application. This phase focuses on code implementation, page construction, interactive feature logic, and system-wide component integration to establish a robust fitness management platform.

02. Objectives of Project Development

Application Implementation

Translate UI/UX designs into operational software logic.

Frontend Development

Build dynamic visual interfaces using modern web technologies.

Feature Implementation

Develop workout management, target setting, and tracking.

Interactive Functionality

Ensure seamless responsiveness across buttons, menus, and controls.

Responsive Design & Layout

Guarantee optimal visual flow across all device form factors.

Code Integration & Debugging

Unify modular sections and proactively handle runtime errors.

03. Frontend Technology Stack

HTML

Provides structural layout, headings, form controls, and component markup.

CSS

Handles styling, responsive layout grids, visual typography, and theme spacing.

JavaScript

Powers dynamic interactions, client-side logic, and event handling.

04. Central Dashboard Development

The dashboard acts as the primary operational hub for FitBuddy users, providing centralized access to core functional areas:

- **Activity Summaries:** High-level visual metrics of daily exercise achievements.
- **Active Workout Plans:** Direct access to scheduled daily training routines.
- **Personal Goals Hub:** Overview of short-term and long-term fitness targets.
- **Activity Records & Progress:** Historical activity logs and performance tracking summaries.

05–07. Core Feature Specifications

FEATURE MODULE	FUNCTIONAL DESCRIPTION & CAPABILITIES
5. Workout Management	Enables users to browse exercise categories, review detailed workout descriptions, follow duration instructions, and customize routines according to goals.
6. Fitness Goals & Tracking	Allows users to configure personal targets, log completed sessions, update metrics, and review historical activity logs for ongoing consistency.
7. Progress Monitoring	Visualizes historical activity records and trends over time using summaries, logs, and chart visualizations to review fitness patterns.

08–09. System Integration & Code Management

DEVELOPMENT PILLAR	IMPLEMENTATION STRATEGY
8. Component Integration	Connects Dashboard, Workout Management, Goal Setting, and Tracking into a unified application linked by a consistent navigation system.
9. Code Management	Maintains structured directory trees using Visual Studio Code for authoring and GitHub for source control, version tracking, and repository backup.

10. Expected Outcome

The project development phase yields a fully functional, user-friendly software build of FitBuddy. It successfully implements workout routines, goal management, tracking tools, and progress analytics into a cohesive application ready for systematic testing.